Horticulture and Food Security and Nutrition: A Systematic Review of Household Level Evidence

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Abstract

Horticulture is widely promoted as a route to better food security and nutrition in low- and middle-income countries, yet the evidence on *whether*, *how*, and *under what conditions* it delivers remains fragmented across crops, interventions, and stages of the value chain. We systematically review 54 empirical studies spanning 21 countries, most in low- and middle-income settings. The evidence most consistently links horticultural interventions to lower food insecurity and improvements in proximal outcomes, including dietary diversity, fruit and vegetable consumption, and food access. Effects on nutritional and anthropometric status are weaker and less consistent, and the direction of effects is more stable than their magnitude. Interventions that combine production support with nutrition education, improved market access, or measures addressing women's time and control tend to produce stronger outcomes than production support alone, although few studies test this contrast directly. The literature focuses primarily on production and availability and income pathways, with little attention to price transmission, market spillovers, or agrobiodiversity. It also concentrates on production-stage interventions, leaving post-harvest, value-chain, and market-level strategies comparatively understudied.

Keywords: horticulture; home gardens; fruits and vegetables; food security; nutrition; dietary diversity; systematic review; low- and middle-income countries.

JEL codes: Q18; I15; O13; Q12.

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1 Introduction

Horticulture—the cultivation of fruits, vegetables, and other crops such as roots and tubers, legumes, herbs, spices, and indigenous vegetables—is a high-value, labor-intensive, and nutrition-sensitive branch of agriculture that is distinct from staple cereal production (FAO, 2020). Horticultural development is increasingly recognized as a key pathway to improving food security and nutrition, particularly in low- and middle-income countries, by increasing the availability of nutrient-dense foods, creating income and employment opportunities along the value chain, and diversifying household production and diets (FAO, 2020; Hiroyuki, 2014). Despite its potential, horticulture remains underrepresented in both agricultural policy and empirical research, which have traditionally focused on staple crops and calorie availability. As a result, access to the diverse and healthy diets that horticulture can provide remains out of reach for many households.

Horticulture differs from the broader agricultural sector in ways that limit the extent to which existing evidence from from staple-crop systems can be generalized. Horticultural production often focuses on crops with high market value and may require more labor and, for some crops, more capital than staple production. In some irrigated settings, horticultural crops generate substantially higher gross margins per hectare than field crops. These features make horticulture an important potential source of both food and income for smallholder farmers (add citation). Fresh fruits and vegetables deteriorate rapidly after harvest, reducing their availability, safety, consumer appeal, and nutritional value. Proper handling, packaging, refrigeration, storage, and market infrastructure are therefore essential for delivering produce to consumers in good condition (add citation). Finally, diversified horticultural systems that include nutrient dense crops can provide nutritional and environmental benefits beyond those of staple dominated production. These effects depend on context and may operate through markets and income rather than through households' consumption of their own production. These characteristics suggest that horticultural interventions may affect food security and nutrition through pathways distinct from those emphasized in the broader agricultural literature.

Despite horticulture's importance for poverty alleviation and food security, and its distinct characteristics relative to other agricultural systems, the existing evidence remains fragmented. Studies typically focus on a single crop, a single intervention—most often home gardens—or a single stage of the value chain, making it difficult to identify the mechanisms through which horticultural interventions affect food security and nutrition. As a result, three questions remain central to policy: whether horticultural interventions improve outcomes, through which mechanisms they operate, and which intervention and policy features most consistently deliver impact.

To address these questions, we conduct a systematic review of 54 empirical studies evaluating horticultural interventions across the value chain. Eligible studies measure at least one food security or nutrition outcome and examine a policy lever or institutional arrangement, including input support, extension, market linkages, nutrition behavior-change communication, and producer organizations.

Horticulture can, in principle, contribute across the six dimensions of food security (HLPE, 2020; Clapp et al., 2022). It raises the *availability* of nutrient-rich foods, improves *access* by generating employment and income along the value chain, supports *utilization* through food

safety and quality standards, strengthens *stability* through diversified production and market participation, enhances *agency*, particularly for women and smallholders who make production and consumption decisions, and can advance environmental *sustainability*. These contributions are not automatic. Healthy, diverse diets built on horticultural crops remain unaffordable for many households in low- and middle-income countries (Bai et al., 2021; Headey and Alderman, 2019), and research and policy have historically prioritized staple-crop self-sufficiency (FAO, 2020; Hiroyuki, 2014). Gender further illustrates this complexity: women's central role in horticultural value chains creates opportunities for empowerment and improved nutrition, but may also increase labor burdens and reinforce unequal control over resources (Van den Broeck and Maertens, 2016).

The multiple, context-dependent pathways linking horticulture to food security and nutrition make it difficult to compare evidence across interventions and settings. To organize this complexity, we adopt a theory of change (Appendix Figure A1, set out in Section 2.1) that distinguishes the mechanisms through which horticultural interventions operate from the proximal and distal outcomes they ultimately affect. The framework links horticultural *interventions* to food security and nutrition *outcomes* through six *mechanisms*: production and availability, income, price and market spillovers, women's empowerment, knowledge and dietary practice, and agrobiodiversity. It distinguishes proximal outcomes, including dietary diversity, food availability, access, and agency, from distal outcomes, including food insecurity, nutritional and anthropometric status, and well-being. Household, market, institutional, program-design, and contextual factors are treated as cross-cutting moderators of intervention effectiveness. We synthesize the evidence within this framework, first assessing the impacts of horticultural interventions, then the mechanisms and moderators that shape those impacts, and finally the intervention and policy features most consistently associated with stronger outcomes.

The 54 included studies span 21 countries and the period 2005–2026, with 87 percent set in low- and middle-income countries and more than half (31 studies) using randomized or quasi-experimental designs that support a causal interpretation. Home and kitchen gardens dominate the intervention literature, and the evidence is concentrated on proximal outcomes—fruit and vegetable consumption and dietary diversity—which are measured far more often than distal nutritional or anthropometric status.

Across this evidence base, we find the clearest and most consistent effects on food insecurity and other proximal outcomes, including dietary diversity, fruit and vegetable consumption, and food access. Effects on more distal nutritional and anthropometric outcomes are weaker and less consistent. Across outcomes, the direction of the effect is more consistent than the magnitude of any individual estimate. Impacts tend to be largest when production support is combined with complementary demand-side or institutional components, such as nutrition education, improved market access, or measures that address women's time constraints and control over resources. However, few studies directly compare these bundled interventions with production support alone, limiting our ability to test this pattern. The evidence focuses overwhelmingly on production-and-availability pathways (51 of 54 studies) and income pathways (42 studies), while other policy levers such as price transmission, market spillovers, and agrobiodiversity remain largely unexamined. The literature also concentrates on the production stage, leaving

post-harvest, value-chain, and market-level strategies comparatively understudied.

The rest of the review is organized as follows. Section 2 sets out the analytical framework, then describes the search, screening, and inclusion process, how mechanisms and outcomes were coded, and the data-extraction and synthesis strategy. Section 3 presents the results, beginning with the characteristics of the included studies and then synthesizing the evidence in three steps: the impacts of horticultural interventions on food security and nutrition, the mechanisms and moderators that shape those impacts, and the intervention and policy features most consistently associated with stronger outcomes. Section 4 discusses the findings and their implications for policy, assesses how much weight the underlying evidence can bear, and sets out the limitations of the review and priorities for future research. Section 5 concludes.

2 Methods and data

We followed a structured, and reproducible systematic-review process. Review procedures—eligibility criteria, the search strategy, and the extraction template—were defined a priori and are reported in full below and in the appendix.

2.1 Analytical framework

The review follows a theory of change specified before data extraction and presented in Appendix Figure A1. The framework links horticultural *interventions* to food-security and nutrition *outcomes* through six *mechanisms*: production and availability, income, prices and market spillovers, women's empowerment, nutrition knowledge and dietary practices, and agrobiodiversity. We also distinguish proximal outcomes, such as dietary diversity, food availability, access, and agency, from distal outcomes, such as food insecurity, nutritional and anthropometric status, and well-being. Household, market, institutional, program-design, and contextual factors may moderate each link in the chain.

The theory of change serves three functions. First, it defines mechanism categories before coding, ensuring consistent classification across studies regardless of their terminology. Second, it distinguishes outcomes from pathways, guiding which measures enter the outcome tables and which are reported as mechanisms. Third, it structures the synthesis: we assess impacts, examine the mechanisms and moderators that shape them, and identify intervention and policy features associated with stronger outcomes.

2.2 Search, screening, and inclusion process

Databases and search strategy: We conducted a structured search of two bibliographic databases—Web of Science and Scopus—to identify empirical studies on horticultural interventions and food security or nutrition outcomes. The search was restricted to work published between 2005 and early 2026; the lower bound was set a priori rather than emerging from the results, so relevant literature published before 2005 is outside the scope of this review.

The search combined two concept blocks with the Boolean `AND`—horticulture, and food security and nutrition—with synonyms within each block joined by `OR` and a third block of excluded topics removed with `AND NOT`. Table 1 reports the terms in each block.

Table 1: Search terms by concept block.

Block	Concept	Terms (combined with `OR`)
1	Horticulture	"horticulture"; horticultural; "horticultural crops"; vegetable*; fruit*; "home garden"/"home gardens"; "kitchen garden"/"kitchen gardens"; "homestead garden"/"homestead gardens"; "homestead production"/"homestead food production"; "school garden"/"school gardens"
combined with `AND`		
2	Food security and nutrition	"food security"; "food insecurity"; nutrition; "nutrition security"; "dietary diversity"; "food availability"; "food access"; "food utilization"; "food stability"; "food sustainability"; "food agency"
combined with `AND NOT`		
3	Excluded topics	"ornamental horticulture"; floriculture; landscaping; "cut flowers"; "food processing"; "food industry"

Note. The same conceptual query was applied across Web of Science and Scopus, with syntax adapted to each database; the executed Scopus `TITLE-ABS-KEY` string is reproduced in Appendix A.2.

Searches were limited to English-language articles and review articles in the social sciences. The core search terms were applied consistently across databases, with syntax adapted to database-specific requirements, including field tags, wildcards, and Boolean operators. The complete search strings are reported in Appendix A.2.

Eligibility criteria:

Studies were screened against the predefined eligibility criteria reported in Table 2. To be included, studies had to meet three criteria: (i) evaluate an intervention with a horticultural production component; (ii) measure at least one food security, nutrition, or well-established intermediate outcome, such as income or food affordability; and (iii) examine a policy-relevant intervention, institutional arrangement, or public or quasi-public policy lever operating at one or more stages of the horticultural value chain. In addition, studies had to report an analytic comparison: an estimate of an association or effect that is identified against a comparison group, a pre-intervention baseline, or a within-sample gradient. Single-arm descriptive surveys, uncontrolled before-and-after evaluations, and post-test-only designs with purposively selected controls were therefore excluded, since they cannot support an estimate of horticulture's contribution to food security. Studies from any geographic setting and population were eligible, although the review prioritizes evidence from low- and middle-income countries.

Screening and inclusion: Records retrieved from the two databases were combined and de-duplicated in Rayyan (Ouzzani et al., 2016), yielding **10,468** unique records. Titles and abstracts were screened in ASReview,[1] with the reviewer applying the predefined eligibility criteria. Screening ended after **1,234** records, or 11.8% of all unique records, had been assessed and the prespecified stopping criterion had been reached. Of the screened records, **82** were selected for full-text review.

Full texts were assessed against the predefined eligibility criteria using the structured

[1] ASReview is an open-source machine-learning tool for systematic reviews. It uses active learning to rank records by predicted relevance, allowing reviewers to examine the most likely relevant records first while retaining full control over inclusion and exclusion decisions (van de Schoot et al., 2021).

Table 2: Eligibility Criteria to Include Studies

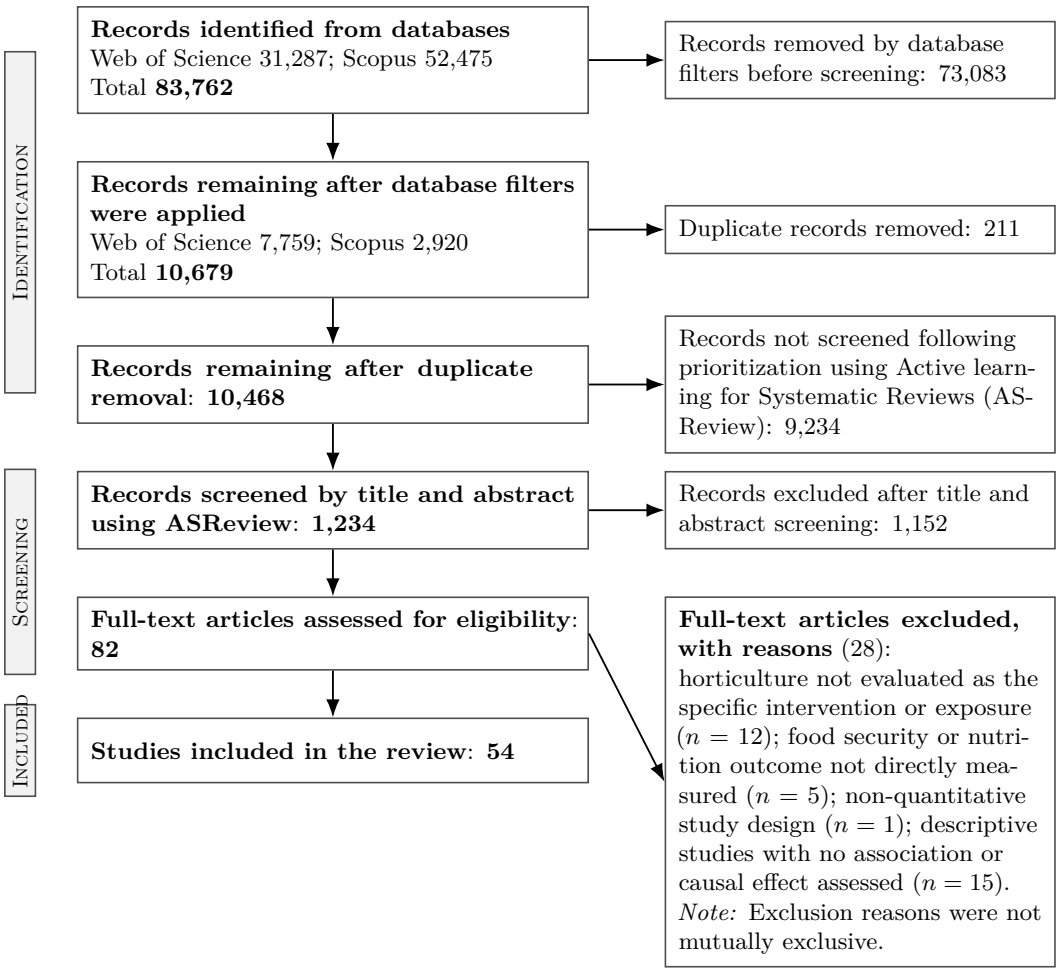
Criterion	**Inclusion**	**Exclusion**
Intervention / exposure	A horticultural production or cultivation component: home, kitchen, school, or community gardens; homestead food production; smallholder vegetable/fruit production; horticultural crops; or indigenous vegetable production	No horticultural production/cultivation component; ornamental horticulture, floriculture, or landscaping; broad agriculture with horticulture incidental (cereals/livestock dominant); food processing or food industry as the primary focus
Outcome	Measures at least one food-security or nutrition outcome (or a well-established intermediate outcome): food availability, food access, dietary diversity, nutrition/anthropometric status, food-insecurity status, food agency, stability; or income/affordability as a pathway	Food security invoked only as background motivation, not a measured outcome; dietary behavior/cooking-skills outcomes only, with no production component
Policy relevance	Examines a policy lever, institutional arrangement, or public/quasi-public intervention operating at one or more value-chain stages (production; post-harvest & distribution; markets & consumption; or cross-cutting), offering actionable insight for policy design, implementation, or evaluation	Purely descriptive or observational studies with no policy lever, institutional arrangement, or public/quasi-public intervention examined
Population	Any (households, children, women, smallholders, communities)	—
Study design	Any empirical design: quantitative, qualitative, mixed methods, impact evaluations, and relevant systematic reviews, provided the study moves beyond description	Non-empirical/opinion pieces; protocol or design papers without outcomes; retracted articles
Setting	Any geographic context (Global South and Global North; LMICs prioritized)	—
Language	English	Non-English
Timeframe	Published 2005–2026	Published before 2005

Note. Studies were required to satisfy *all three* of the first criteria—a horticultural component, a food-security/nutrition (or intermediate) outcome, and policy relevance—consistent with the review's focus on *which interventions and policies* promote and sustain horticultural development for improved food security (RQ3).

extraction template described below. Of the **82** full texts assessed for eligibility, **54** met all inclusion criteria. The remaining **28** were excluded because horticulture was not evaluated as the specific intervention or exposure ($n = 12$), no food-security or nutrition outcome was directly measured ($n = 5$), the study design was non-quantitative ($n = 1$), or the study was descriptive with no association or causal effect assessed ($n = 15$). These reasons are not mutually exclusive, so they sum to more than 28.

A single reviewer conducted the initial screening and data extraction using the active-learning workflow. To strengthen the reliability of these decisions, a senior reviewer independently reassessed borderline exclusions. All included studies were also checked against the original full texts as part of a structured quality-assurance process. Figure 1 summarizes the complete screening process and the reasons for exclusion at the full-text stage.

Figure 1: Study identification, screening, and inclusion process



Quality appraisal: Each included study was appraised against the established instrument matched to its design family, rather than against a single generic checklist. Randomized and cluster-randomized trials were appraised with RoB 2 across its five domains (Sterne et al., 2019); non-randomized studies of interventions—including difference-in-differences, propensity-score matching, endogenous switching regression, and instrumental-variable designs—with ROBINS-I across its seven domains (Sterne et al., 2016); analytical cross-sectional studies with the eight-

item JBI checklist (Moola et al., 2020); mixed-methods studies with the mixed-methods category of the MMAT (Hong et al., 2018); and systematic reviews with AMSTAR-2 (Shea et al., 2017). Design-matched instruments were used because a generic checklist cannot interrogate what distinguishes these designs: ROBINS-I asks directly about the confounding-adjustment strategy and the untested identifying assumptions—parallel trends, exclusion restrictions—that carry the weight in econometric evaluations, and the MMAT alone assesses whether the qualitative and quantitative components were genuinely integrated.

Each instrument is applied with its own domains and its own decision rule, and every domain judgement is recorded with a verbatim supporting quotation located in the source article. The full appraisal—each study, its instrument, its per-domain judgements, and the justification for each judgement—is reported in the supplementary quality-appraisal workbook.

[TODO: Quality appraisal results pending. This subsection states the instruments only. Still to be settled and then written up here and in the Discussion: how each instrument's overall judgement is derived, whether ratings from the five instruments are reported natively or mapped to a common scale, and the resulting distribution across the included studies. No rating counts appear anywhere in the manuscript until that is fixed.]

2.3 Data extraction

We extracted data from each of the 54 included studies using a standardized template, reproduced in Appendix A.3. The template captured bibliographic information, study setting and design, sample characteristics, the horticultural intervention, food-security and nutrition outcomes, underlying mechanisms, determinants and effect modifiers, heterogeneity, intervention and policy features, study quality, key findings, value-chain stage, and the policy lever or institutional arrangement examined.

A single reviewer conducted the initial extraction. Each record identified the reviewer and the date of extraction. As part of the structured quality-assurance process, all extracted information was subsequently checked against the original full texts.

2.4 Coding of mechanisms and outcomes

The theory of change in Figure A1 supplies the categories the synthesis is reported against. We code six impact pathways—production and availability, income, knowledge and dietary practice, women's empowerment, agrobiodiversity, and price and market spillovers—and record, for each study, which of them it presents evidence on. Categories are not mutually exclusive: a study commonly reports several, so pathway counts sum to more than the number of studies. Because primary studies do not describe mechanisms in a standardized way, and few label them as such, pathway assignment rests on keyword coding of the extracted mechanism text rather than on any field the original authors completed. The resulting counts are therefore best read as indicators of the relative prominence of each pathway—the ranking is more informative than any individual figure—rather than as precise prevalence rates.

One boundary is worth stating explicitly, because it determines what is counted as an outcome and what is counted as a pathway. We reserve the outcome category for food-security and

nutrition outcomes proper: availability, access, dietary diversity, nutritional and anthropometric status, anemia, and food agency. Income, poverty, production volumes, assets, labour, and nutritional knowledge are treated as pathway variables and reported with the mechanisms, even where a study's own abstract presents them as outcomes. The distinction matters because income in particular is reported as a headline result in much of this literature, and counting it as a food-security outcome would make the evidence base appear to speak to consumption more directly than it does. Routing a variable to the pathway side never discards its evidence: the corresponding estimates are extracted and reported alongside the mechanism they belong to.

2.5 Evidence synthesis

Variation in interventions, study designs, outcome definitions, and measurement approaches precluded a meaningful meta-analysis. We therefore conducted a structured narrative and thematic synthesis guided by the review's theory of change (Figure A1). The synthesis addresses three questions: (i) how effects vary across outcome domains and intervention types (RQ1); (ii) which mechanisms and contextual factors help explain this variation (RQ2); and (iii) which intervention and policy features are consistently associated with better outcomes (RQ3).

3 Results

3.1 Characteristics of the included studies

The searches of Web of Science and Scopus yielded 10,468 unique records after de-duplication in Rayyan. Using ASReview, we manually screened 1,234 records, of which 82 were retained for full-text assessment. Of these, **54** met all eligibility criteria and were included in the review (Figure 1). The included studies were published between 2005 and 2026 and cover **21 countries**. Table A5 reports their full study-level characteristics. The remainder of this subsection describes their distribution across income settings, study designs, intervention types, outcomes, and mechanisms. **Geographic distribution:** The evidence comes predominantly from low- and middle-income countries (LMICs), which account for **47 of 54 studies (87%)**. Five studies (9%) focus on high-income countries, and two (4%) cover both income groups (Table 3). Kenya is the most frequently studied country ($n = 6$), followed by the United States ($n = 5$) and Bangladesh ($n = 5$), then Ethiopia ($n = 4$). The LMIC evidence is therefore concentrated in Sub-Saharan Africa and South and Southeast Asia.

Table 3: Included studies by income setting.

Income setting	**Studies (n)**	**%**
LMIC	47	87%
HIC	5	9%
Mixed (LMIC and HIC)	2	4%

Study design and identification: Quantitative studies account for 44 of 54 studies in the

review. These include 10 randomized experiments, 21 quasi-experimental studies, 12 observational studies, and one economic evaluation. The quasi-experimental studies use methods such as difference-in-differences, instrumental variables, endogenous switching regression, and propensity-score matching. The remaining studies comprise 6 mixed-methods studies and four systematic reviews (Table 4). Together, randomized and quasi-experimental studies represent 31 studies, or 57% of the evidence base. These designs provide the strongest basis for causal inference, although the credibility of identification varies across studies. Most of the remaining evidence documents associations rather than causal effects.

Table 4: Included studies by study design and identification strategy.

Study design	**Studies (*n*)**	**%**
Quantitative	44	81%
Experimental (RCT)	10	19%
Quasi-experimental	21	39%
Observational	12	22%
Economic evaluation	1	2%
Mixed methods	6	11%
Systematic review	4	7%

Note. Quantitative studies are subdivided by the design recorded for each study: experimental (randomized), quasi-experimental (DiD, IV, ESR, or propensity-score matching), observational (correlational or descriptive), and economic evaluation. Percentages are of the 54 included studies.

Intervention types: Home and kitchen gardens are the most frequently studied intervention ($n = 25$). Other common interventions combine horticultural production with nutrition education or behavior-change communication ($n = 15$), irrigation or water-management support ($n = 8$), indigenous-vegetable production ($n = 7$), smallholder fruit and vegetable production ($n = 6$), or community gardens ($n = 5$). Far fewer studies examine value-chain or market-linkage interventions ($n = 5$) and school gardens ($n = 3$) (Table A2).

Outcomes measured: The literature focuses primarily on proximal food-security and dietary outcomes. Fruit and vegetable consumption is the most frequently measured outcome ($n = 31$), followed by dietary diversity, commonly measured using the HDDS or MDD-W ($n = 31$). Studies also examine food insecurity using measures such as the HFIAS, FIES, or FCS ($n = 21$), nutritional or anthropometric status ($n = 20$), and food availability or access ($n = 13$) (Table A3). Following the theory of change, we classify income as a mechanism rather than as a food-security or nutrition outcome and report it with the other pathways below. Outcome categories are not mutually exclusive, and most studies measure more than one outcome. The literature emphasizes proximal outcomes, such as dietary diversity and consumption, more heavily than distal outcomes, such as anthropometric status.

Mechanisms discussed: Nearly all studies discuss the production and food-availability

pathway ($n = 51$). The income pathway is also prominent ($n = 42$), followed by changes in nutrition knowledge and dietary practices ($n = 29$). Fourteen studies discuss women's empowerment, whereas only two examine agrobiodiversity and one examines prices or market spillovers (Table A4). These categories are based on keyword coding of the extracted text because the primary studies do not report mechanisms in a standardized manner. The counts should therefore be interpreted as indicators of the relative prominence of each pathway rather than as precise measures of prevalence. The ranking of the pathways is more informative than any individual count. Figure 2 summarizes the distributions of studies across interventions, outcomes, and mechanisms.

Figure 2: Distribution of included studies by intervention, outcome, and mechanism



Note: Panels report the number of included studies by (a) horticultural intervention type, (b) food-security and nutrition outcome measured, and (c) impact pathway discussed. Categories are not mutually exclusive, so the counts within each panel may sum to more than 54. Appendix A.4 reports the complete counts.

3.2 The impact of horticulture on food security and nutrition

The literature linking horticulture to food security and nutrition is substantial, but its ability to identify causal effects varies considerably across studies. Of the 54 studies reviewed, 31 use designs intended to support causal inference, including randomized controlled trials, difference-

in-differences, propensity-score matching, instrumental variables, and endogenous switching regression. The remaining 23 rely on correlational or descriptive analyses, use mixed-methods designs, or are systematic reviews. The four systematic reviews are narrative syntheses rather than meta-analyses: none pools effect sizes across its included studies (Garcia et al., 2018; Burt et al., 2021; Ndwandwe et al., 2026b,a), and three state explicitly that the evidence they summarize cannot establish causality (Garcia et al., 2018; Burt et al., 2021; Ndwandwe et al., 2026a).

Research attention is concentrated on proximal outcomes. Fruit and vegetable consumption is examined in 31 studies and dietary diversity in 31, followed by food insecurity ($n = 21$), nutritional or anthropometric status ($n = 20$), and food availability and access ($n = 13$). These categories are not mutually exclusive because many studies examine more than one outcome. Following the coding framework described in Section 2.4, we classify income as a pathway rather than an outcome and discuss it below. Overall, the evidence base is densest for outcomes closest to production and consumption and becomes progressively thinner for distal outcomes such as nutritional and anthropometric status.

Dietary diversity has the strongest and most consistent evidence base. Experimental and quasi-experimental evaluations generally find positive, and often substantial, effects of horticultural interventions. Home-garden adoption increases household dietary diversity by about 10 percent in Odisha (Ogutu et al., 2023) and by 13 points on a frequency-weighted index ranging from 208 to 570 in coastal Bangladesh (Rahman et al., 2024). In Ghana, correcting for selection into climate-smart gardening yields an estimated gain of 37.5 percent, although the unadjusted difference between adopters and non-adopters is not statistically significant (Prah et al., 2026). Correcting for selection similarly yields a gain of nearly 38 percent among guava adopters in Tigray (Kahsay et al., 2025), while effects are particularly large among land-constrained households in Rwanda (Issahaku et al., 2023). Market-oriented horticultural production also improves dietary diversity, although estimated effects tend to be smaller (Kone et al., 2025; Mossie and Elias, 2025). A panel evaluation in Dhaka finds a modest 0.37-point increase in women's dietary diversity ($p = 0.032$), while a multicountry trial of homestead food production finds limited effects that vary across countries (Schreinemachers et al., 2025; Depenbusch et al., 2021). A cluster-randomized home-garden and nutrition program in Cambodia raised the quantity of vegetables households consumed by 60.8 grams per person per day ($p < 0.01$) but left household, women's, and children's dietary diversity scores unchanged (Depenbusch et al., 2022). Taken together, the evidence is broadly positive, although effect sizes vary considerably across interventions and settings.

Observational evidence points in the same direction but also reveals disparities within households. In Ugandan home gardens, plant-species richness is positively associated with both household and women's dietary diversity. Yet the median household score exceeds its adequacy threshold (6 versus 5.9), whereas the median scores for women (4.5 versus 5) and toddlers (3.5 versus 4) remain below their respective thresholds (Whitney et al., 2018). Household-level measures may therefore overstate improvements in diet quality among nutritionally vulnerable household members. What a diversity score captures also matters. In Nairobi's Kibera settlement, households practicing sack gardening consumed a wider range of vegetables than non-gardening households (10.6 versus 9.2 items, $p < 0.001$), yet the two groups recorded the same overall

dietary diversity (6.4 food groups each), because the garden produce fell within a food group most households already consumed (Gallaher et al., 2013).

Evidence on fruit and vegetable (FV) consumption likewise favors interventions that combine production support with demand-side measures. Experimental and quasi-experimental evaluations generally find that gardening increases FV intake when paired with nutrition education, school feeding, or food distribution. In Texas, a school gardening and nutrition program increased vegetable intake by 0.48 servings per day, compared with 0.04 servings in the control group ($p = 0.02$) (Davis et al., 2021). A California intervention combining school gardens with salad bars increased vegetable consumption by 56 percent (Taylor et al., 2018). Evidence from a cluster-randomized trial in Alabama further indicates that nutrition education, rather than gardening alone, drove the improvement (Sanchez et al., 2025). Similarly, nutrition and production training in Bangladesh increased per capita vegetable consumption by 19.3 grams per day (Schreinemachers et al., 2016).

Interventions focused on production alone yield less consistent results. A school-garden trial in Nepal improved nutrition knowledge but did not increase consumption (Schreinemachers et al., 2017). A cluster-randomized school-garden trial in Burkina Faso that paired cultivation with agriculture, nutrition, and hygiene education likewise raised children's knowledge without changing intake: the pooled effect on the share of children who ate any vegetable the previous day was 11.1 percentage points and not statistically significant ($p = 0.186$) (Schreinemachers et al., 2019). The trial was underpowered for this outcome—its detectable difference was 19 percentage points—and the authors attribute the absence of behavior change to poor water and sanitation facilities and to garden supply concentrated in the three to four months when water was available. Observational comparisons generally associate gardening with higher FV intake, but self-selection into gardening limits a causal interpretation of these differences (Bushamuka et al., 2005). Three years into a kitchen-garden and nutrition-education program in the Nepali terai, participant households reported eating the home-produced crop more often than non-participants for 14 of the 16 promoted vegetables and fruits and were more likely to have preserved food in the previous year (86.9 versus 60.1 percent, $p < 0.005$), although the outcomes were self-reported by respondents trained on the behaviors being measured and the comparison has no baseline (Jones et al., 2005). A systematic review of twelve studies of adult urban gardeners reports the same pattern, with participation consistently associated with greater fruit and vegetable consumption, a more varied diet, and improved perceived access to fresh and healthy food, although the review synthesizes no effect sizes (Garcia et al., 2018). Overall, the evidence suggests that increasing horticultural production may not be sufficient to change diets unless interventions also address nutrition knowledge, food access, or other constraints on demand.

Evidence on food insecurity is generally consistent in direction, although the largest estimates emerge after correcting for non-random adoption and therefore depend on the underlying identifying assumptions. Across studies using the Household Food Insecurity Access Scale (HFIAS), Food Insecurity Experience Scale (FIES), and Food Consumption Score, horticultural interventions generally improve food security. In Ghana, commercializing home-garden production substantially reduces HFIAS scores (Nkomoki et al., 2025). In Lesotho, correcting for selection into keyhole-garden adoption yields an increase of more than 30 percentage points in

the probability of being food secure and a decline of over 40 percentage points in severe food insecurity (Ogunleye et al., 2025). The distributional shift is uneven, however: many households move from severe to moderate food insecurity rather than becoming food secure. Dual-purpose sweetpotato adoption in Rwanda similarly reduces severe food insecurity by 19 percentage points (Danso-Abbeam et al., 2021), while a vegetable-kit and irrigation package in Syria increases Food Consumption Scores by 21 percent (Baliki et al., 2024).

Other studies find smaller or no improvements. In Limpopo, South Africa, an endogenous switching model of 2,043 rural households likewise reports lower average household food insecurity with home- garden participation, but the effect for the households that actually garden is not statistically significant ($p = 0.712$) and the reported treatment-effect magnitudes lie far outside the 0–27 range of the HFIAS they are said to act on (Gwacela et al., 2024). In southern Ethiopia, matching Moringa-producing households to comparable non- producers associates production with Food Consumption Scores 4.7 to 6.2 points higher, consistently across three matching algorithms (Tafesse et al., 2020). After three years of participating in a homestead gardening and broiler program in KwaZulu-Natal, most households remain severely food insecure despite reporting improvements in perceived diet quality (D'Haese et al., 2013). Thus, perceived dietary improvements do not necessarily translate into measurable gains on validated food-security indices. A systematic review of 31 community-garden studies documents the same divergence: participants perceived a gain in food security in eight of the twelve studies that addressed it, yet of the three studies that measured food security, two found no statistically significant benefit (Burt et al., 2021).

Nutritional status and anthropometric outcomes remain the least studied and show the weakest response to horticultural interventions. The strongest evidence comes from evaluations of enhanced homestead food production. In Nepal, the intervention reduced anemia among children and mothers (Osei et al., 2017). In Burkina Faso, it reduced the prevalence of underweight among women by 16.4 percentage points, although the estimate was only marginally statistically significant ($p = 0.09$) (Bliznashka et al., 2022). The treatment–control difference persisted two years after the program ended, but this follow-up estimate is not causally identified because it relies on a small, non-randomly retained subsample. Evidence beyond these studies is limited. Studies in Rwanda and Texas find modest or no changes in body mass index (Issahaku et al., 2023; Davis et al., 2021). In Cambodia, households in a homestead food production program had higher household dietary diversity than controls (4.3 versus 3.8, $p < 0.05$) and consumed micronutrient-rich foods more often, yet differed on none of the nutrition-status measures, including child stunting (40.5 versus 42.3 percent), underweight, wasting, hemoglobin, and anemia (Olney et al., 2009). The authors caution that baseline non-comparability, self-selection into the program, and baseline and endline surveys falling in different seasons preclude attributing any of these differences to the program. Overall, horticultural interventions appear more effective at improving dietary diversity and food consumption than nutritional status or anthropometric measures. These downstream outcomes may require longer exposure and depend on health, sanitation, disease, and other factors beyond diet alone.

Evidence on food availability and access is mostly descriptive, with few studies providing causal estimates. The available research nevertheless indicates that horticulture expands the

local supply of diverse foods through household production and may improve access by strengthening market participation. These benefits depend on access to complementary resources and institutions. In northern Ethiopia, irrigation access and cooperative membership are jointly associated with greater participation in vegetable markets and improved food availability (Gebru et al., 2019).

This review treats income as a pathway to food security rather than as a food-security outcome. It is also the pathway for which the evidence is most consistent, although most studies rely on cross-sectional data and attempt to correct for non-random adoption. In Kenya, commercial production of African indigenous vegetables increases per capita income (Krause et al., 2019), while adopting tissue-culture bananas more than doubles farm income (Kabunga et al., 2014). Guava adoption generates similarly large income gains in Ethiopia (Kahsay et al., 2025). The benefits also vary across households: female-headed households gain more income from home gardening in Odisha (Ogutu et al., 2023).

Higher income, however, does not necessarily improve nutrition. In western Kenya, commercializing indigenous vegetables increased non-food expenditure but did not improve nutritional outcomes. Farmers expanded production for sale rather than diverting output from markets toward household consumption (Kubitza et al., 2024). Income growth may therefore be insufficient to improve diets when market incentives favor sales and interventions do not also address household consumption and nutrition.

3.3 Mechanisms and moderating factors

Production and food availability constitute the most frequently documented mechanism, appearing in 51 of the 54 studies reviewed. The pathway is direct: horticultural interventions increase the quantity and diversity of food that households produce, expanding the food available for their own consumption and generating marketable surpluses. Evidence for this initial step is strong and consistent. In Bangladesh, home-garden training increased annual per capita vegetable harvests by 86 percent in the year after training (Schreinemachers et al., 2015). Another homestead-production program generated median harvests almost three times larger among participant households than among controls (135 versus 46 kg) (Bushamuka et al., 2005). Production typically responds quickly, can be measured directly, and therefore provides the clearest link between horticultural interventions and subsequent food-security and nutrition outcomes. Production gains need not accrue to the most food-insecure households, however: across 18 community gardens on the California coast, self-reported output is higher among gardeners with higher incomes and more hours spent gardening, whereas food insecurity is concentrated among gardeners with lower incomes and larger families (Lin et al., 2024).

Income is the second most frequently documented pathway, appearing in 42 studies, and operates mainly through the commercialization of horticultural production. By increasing sales, horticultural interventions can raise household purchasing power for both food and non-food goods. Studies document income gains from commercializing indigenous vegetables and adopting tissue-culture bananas in Kenya, as well as from adopting guava in Ethiopia (Krause et al., 2019; Kubitza et al., 2024; Kabunga et al., 2014; Kahsay et al., 2025). In the tissue-culture banana study, however, the estimated gain emerges only after correcting for negative selection into

adoption; neither the raw comparison nor the OLS estimate indicates a statistically significant difference.

Income gains do not necessarily translate into better diets. Selling horticultural products can compete with consuming them at home, weakening the connection between greater income and improved nutrition (Kubitza et al., 2024; Mossie and Elias, 2025). Accordingly, estimated effects on food security tend to be smaller and less precise than the corresponding effects on income.

Knowledge and dietary practices appear to improve most when nutrition behavior-change communication (BCC) is combined with production support. Across the 29 studies examining this pathway, bundled interventions generally perform better than either component alone, although no study in this review tests that comparison within a single experimental design. In Missouri, participants who received both nutrition education and produce from community gardens reported the highest odds of meeting recommended fruit and vegetable intake levels (Barnidge et al., 2014).

Greater knowledge alone does not necessarily change diets. A school-garden trial in Nepal substantially improved nutrition knowledge but had no measurable effect on fruit and vegetable consumption or children's height-for-age (Schreinemachers et al., 2017). The same break appears in Burkina Faso, where a school garden with complementary agriculture, nutrition, and hygiene lessons raised children's knowledge of food and nutrition by 6.6 percentage points ($p = 0.046$) and of sustainable agriculture by 5.0 percentage points ($p = 0.037$), yet moved neither their awareness of fruits and vegetables nor their stated food preferences nor their reported intake (Schreinemachers et al., 2019). These findings suggest that nutrition education is more likely to change dietary behavior when households also have the physical and economic means to obtain healthy foods.

Women's empowerment appears as a potential mechanism in 14 studies, reflecting women's central roles in horticultural production, marketing, and household food decisions. Some studies associate greater control over production, income, or decision-making with improved food security and nutrition, but few test this pathway directly. In Burkina Faso, an intervention delivered through female-led village model farms reduced maternal underweight but did not improve any measured dimension of women's empowerment; it also slightly reduced women's meetings with other women (Bliznashka et al., 2022). In Bangladesh, training increased women's participation in gardening, sales, and control over revenue alongside greater diversity in the vegetables consumed (Schreinemachers et al., 2015). Yet annual sales revenue remained below one dollar, and measures of food access did not improve. These effects are not automatic. Commercialization can increase women's workloads and reinforce unpaid labor burdens. Evidence from peri-urban Botswana therefore suggests that greater participation in horticulture improves women's agency and household food security only when interventions also address the distribution of labor and control over the resulting income (Fehr and Moseley, 2019). A cluster-randomized home-garden program in Cambodia gives a direct estimate of that trade-off: it increased women's time in the garden by 29 minutes per day ($p < 0.01$) without improving their empowerment score (Depenbusch et al., 2022). Overall, women's empowerment is a plausible pathway, but the available evidence does not yet establish it as a consistent mediator of horticulture's effects on food security and nutrition.

Agrobiodiversity and market spillovers remain major evidence gaps. Only two studies explicitly examine agrobiodiversity. In Yucatán, plant-species diversity is associated with higher Food Consumption Scores (Castañeda-Navarrete, 2021). In Uganda, greater plant-species richness in home gardens is associated with higher dietary diversity among households and women (Whitney et al., 2018). The Ugandan study also describes crop diversity as protection against lean-season shortages, although it does not estimate this buffering effect. The small evidence base contrasts with the prominent role assigned to agrobiodiversity in nutrition-sensitive agriculture.

Evidence on broader market effects is scarcer still. No study in this review estimates how horticultural interventions affect prices, local markets, or households beyond direct participants; the literature's empirical focus remains confined to production and consumption within participating households. General-equilibrium and spillover effects are therefore an open question rather than a contested one. One evaluation does report such spillovers without estimating them: in the Marshall Islands, participants in a horticulture aid project scored no higher than non-participants on nutrition knowledge or attitudes, a null the authors attribute to knowledge being shared with neighboring households (Cheng et al., 2021).

Differences in household resources, market conditions, institutional support, and program design help explain the variation in impacts across studies. Land and water are particularly important. Land-constrained households in Rwanda benefit disproportionately from horticultural interventions, while affordable water in Botswana and irrigation access in Ethiopia help translate production into food-security gains (Issahaku et al., 2023; Fehr and Moseley, 2019; Gebru et al., 2019; Kahsay et al., 2025). Market access and seasonality also condition the effects. In rural Myanmar, by contrast, the association between home gardening and lower food insecurity strengthens as landholding rises: interactions between garden access and land size are significant for all five food-insecurity indicators, and the marginal association with having to change the diet grows from -7.2 to -19.6 percentage points between the smallest and the largest land category (Rammohan et al., 2019). Dietary diversity improves more in areas with better market access in Côte d'Ivoire, whereas the food supplied by South African home gardens varies substantially across seasons (Kone et al., 2025; Malatsi et al., 2023). A review of 53 reports on indigenous leafy vegetables in Sub-Saharan Africa likewise attributes constrained cultivation to climate change and rainfall variability, pests, land degradation and water scarcity, insecure land tenure, limited training, and restricted market access, compounded by a social stigma that these crops are "food for the poor" (Ndwandwe et al., 2026b).

Institutional support appears to influence participation more clearly than final outcomes. Extension contact, group membership, and access to credit predict the *uptake* of climate-smart home gardening in Ghana, while extension access predicts participation in indigenous-crop markets in South Africa (Prah et al., 2026; Zondi et al., 2022). Evidence on program design is more qualified. Combining production support with nutrition behavior-change communication improves nutrition knowledge and dietary diversity relative to no intervention (Schreinemachers et al., 2015), but no study in this review compares such a program with production support alone, so the marginal contribution of the behavior-change component is untested. Moreover, both bundled and stand-alone production interventions often leave food-insecurity and anthropometric

outcomes unchanged (Schreinemachers et al., 2015; Davis et al., 2021). Agroecological and sociocultural conditions introduce further variation across settings and populations (Keding et al., 2012; Castañeda-Navarrete, 2021).

3.4 Effective interventions and policy features

Consistent with the descriptive findings, most interventions and policy levers operate at the production stage. These measures include seed kits, tissue-culture planting material, irrigation, container gardening, and soil-water-retention technologies. Input and technology support is generally associated with better outcomes, although studies rarely separate its effects from those of accompanying training. In Syria, a package combining vegetable kits with drip irrigation improved food security (Baliki et al., 2024). In dry-season Ghana, 12 percent of women receiving container-gardening and soil-water-retention support had non-diverse diets at endline, compared with 62 percent in the control community (Kubuga, 2025). Another Ghanaian study finds that water availability increases the market orientation of home gardens, while access to input subsidies directly improves food security (Nkomoki et al., 2025).

Extension and training accompany most production interventions, including tissue-culture banana propagation and integrated pest management for mango in Kenya (Kabunga et al., 2014; Nyang'au et al., 2020). Several studies attribute limited adoption to insufficient extension contact rather than a lack of farmer interest. In Ghana, for example, a simulation indicates that expanding access to extension services and social groups by 15 percent would approximately double home-garden adoption (Prah et al., 2026). This result predicts the effects of expanded access, however, rather than evaluating an implemented extension policy.

Post-harvest and distribution interventions are almost entirely absent: no included study evaluates a storage, aggregation, or distribution intervention as its primary lever. This is difficult to reconcile with the documented importance of post-harvest losses. In Ethiopia, for example, such losses account for more than one-quarter of apple production (Mossie and Elias, 2025). The scarcity of interventions targeting storage, aggregation, and distribution therefore represents a gap in the policy evidence.

Market-linkage and demand-side interventions are also underrepresented, and much of the available evidence is associational. In Côte d'Ivoire, proximity to markets and access to market-extension services are associated with higher Food Consumption Scores and greater caloric intake among market-oriented farmers (Kone et al., 2025). In western Kenya, cooperative membership is associated with greater sales of indigenous vegetables and higher non-food expenditure (Kubitza et al., 2024). Demand-side behavior-change communication (BCC), combined with school procurement, also increased vegetable selection in California (Taylor et al., 2018). More broadly, studies identify cooperatives and producer groups as potentially important design features, including indigenous-vegetable cooperatives in Ghana and savings-and-credit cooperatives supporting irrigation-based vegetable businesses in Ethiopia (Ayanwale et al., 2025; Gebru et al., 2019). A review of 38 studies of indigenous vegetables grown in Sub-Saharan African home gardens matches the strategies farmers use against the risks they face— improved and stress-tolerant varieties, cooperatives, training, storage, and value addition—but evaluates none of them (Ndwandwe et al., 2026a).

Cross-cutting interventions that combine production support with demand-side or institutional measures are promising, but their incremental effects remain uncertain. School-garden programs that pair cultivation with nutrition curricula and school-meal procurement produce some of the clearest consumption gains, although these gains are generally modest (Taylor et al., 2018). In a large Alabama trial, adding a garden to nutrition education generated no additional benefit; the interaction between education and gardening was negative for the change in vegetable consumption (Sanchez et al., 2025). The review's single economic evaluation costs one such bundle. An enhanced homestead food production program in Cambodia that combined home gardens with poultry and aquaculture raised children's usual dietary zinc intake by 12 percent and vitamin A intake by 30 percent and returned a benefit–cost ratio of 1.62 over ten years, but the monetized value of averted micronutrient disease accounted for only 4 percent of total benefits, so the economic case rests mainly on the increased value of household production (Dragojlovic et al., 2020).

Evidence on whether effects persist after interventions end is mixed. In a three-year Tanzanian trial, cultivation persisted: significantly more households continued to grow crops after 36 months, even though the earlier gain in dietary diversity had disappeared (Blakstad et al., 2022). In Burkina Faso, the treatment–control difference in women's underweight prevalence remained two years after the intervention ended (four-year difference-in-differences estimate: 18.3 percentage points, $p = 0.02$) (Bliznashka et al., 2022). This post-program estimate is not causally identified, however, because it relies on a small, non-randomly retained subsample.

Overall, the value-chain mapping confirms that evidence on effective policy instruments is concentrated at the production stage. Far less is known about post-harvest, distribution, and market interventions, even though income generation, market spillovers, and post-harvest losses extend well beyond the farm gate.

4 Discussion

Three patterns emerge from the synthesis. First, estimated effects follow a proximal-to-distal gradient: evidence is strongest for dietary diversity, fruit and vegetable consumption, and household food security, but much weaker for nutritional status and anthropometric outcomes. Second, outcomes depend not only on horticultural production itself but also on the complementary support with which it is delivered. Third, research is concentrated on household production and income, leaving agrobiodiversity, prices, and wider market spillovers largely unexplored.

The proximal-to-distal gradient may reflect differences in measurement and timing rather than a limit to horticulture's nutritional potential. Production, income, and diets can respond within a season, whereas growth and biomarker outcomes develop over longer periods and depend on health, sanitation, disease, and other non-food inputs. Evaluations conducted over short periods and focused on farm-level outcomes are therefore more likely to detect changes in diets than in anthropometric status. The appropriate policy conclusion is not that horticulture cannot improve nutrition, but that its downstream effects require longer follow-up periods and a broader set of outcomes to assess credibly.

The results also show why production gains alone may not improve diets. Greater output

does not ensure that households consume the additional produce rather than sell it, that produce reaches markets before spoiling, or that additional income is spent on nutritious food. Nutrition behavior-change communication, market linkages, and women's control over income each address a different constraint along this pathway. Bundled programs may therefore be more effective than production support alone, although few studies directly compare the two. The recurring finding that commercialization raises income without improving nutrition illustrates the same point: households may expand production for sale without increasing their own consumption. Income is thus a potential pathway to better diets, but its nutritional effect depends on how households allocate produce, earnings, and decision-making authority.

The substantial heterogeneity across studies further indicates that context conditions the returns to horticultural interventions. Benefits tend to be larger among land- and resource-constrained households and in settings with affordable water, accessible markets, extension services, cooperatives, and credit. Effects weaken when these complementary conditions are absent. Horticultural interventions are therefore better understood as a family of context-dependent programs than as a uniform treatment. This perspective also underscores the importance of the sparsely studied agrobiodiversity and market- spillover pathways. Because these mechanisms operate beyond participating households, household-level evaluations reveal little about horticulture's aggregate or general-equilibrium effects.

Policy implications

1. **Design beyond production alone.** Pair production support with complementary measures that address relevant constraints, such as nutrition behavior-change communication or market linkages. The current evidence favors such bundles, although their incremental effects require more direct testing.
2. **Match program design to the intended outcome.** Programs targeting diet quality should address nutrition knowledge and access to nutritious foods. Those targeting income or household food security should strengthen market access and producer organizations. Anthropometric objectives require longer implementation and evaluation periods than most existing programs provide.
3. **Support women's agency without increasing unpaid work.** Programs should strengthen women's control over production and income while monitoring how horticultural activities affect their time and workload.
4. **Invest beyond the farm gate.** Storage, aggregation, distribution, and market-linkage interventions receive little attention despite substantial documented post-harvest losses. These stages warrant greater policy experimentation and evaluation.

The strength of the evidence base

The conclusions must be weighed against substantial limitations in the underlying evidence. Only 10 of the 54 included studies (19 percent) use randomized designs, and 31 (57 percent) use designs capable of supporting a causal claim. Nine studies (17 percent) have no comparison group. **[TODO: quality-appraisal distribution pending — see the Methods note]** The

literature also emphasizes proximal outcomes over nutritional status, while variation in outcome definitions and measurement instruments precludes meta-analysis and limits comparisons of effect sizes.

This evidence base is weak relative to some adjacent literatures. An evidence gap map covering 1,838 evaluations of food-system interventions in low- and middle-income countries reports that 75 percent use randomization as their main identification strategy (Storhaug et al., 2022). Because that review includes only experimental and quasi-experimental evaluations, the appropriate comparison is with the 31 studies in our review that use causal-capable designs. Of these, 10—or 32 percent—are randomized, still well below the corresponding share in the evidence gap map. Weak research designs are not unique to horticulture: only 15 of 78 studies in a review of farm-level livestock interventions use randomized designs (Lindahl et al., 2020), and concerns about the evidence linking agriculture to nutrition are long-standing (Ruel et al., 2018). More than a decade ago, a systematic review concluded that the effect of agricultural interventions on child nutrition could not be established confidently because of methodological weaknesses (Masset et al., 2012). The design profile observed here shows that this concern remains relevant to the horticultural literature.

Our source-verification process also permits a direct assessment of reporting quality. We traced every quantitative value to a verbatim passage or table and recorded its location in the source. Of the 54 studies, 49 (91 percent) contain at least one documented reporting problem, producing 96 issues in total. These include significance markers inconsistent with reported standard errors, effect estimates that cannot be reproduced using the method described, analytic sample sizes recoverable only by back-solving reported statistics, reversed sign conventions, and totals outside the range implied by their components. We recorded a further 41 unresolved questions across 26 studies. Whenever the text and tables disagreed, we used the tabulated result and documented the discrepancy.

Two conclusions follow. First, the direction of the evidence is more credible than the precise magnitude of the reported effects. Estimates for proximal outcomes are generally positive, but many effect sizes depend on corrections for non-random selection, small analytic samples, or single cross-sections. Second, the principal obstacle to quantitative synthesis is not simply the number of available studies. Inconsistent measurement and incomplete or irreproducible reporting also prevent reliable meta-analysis.

Strengths and limitations

This review covers 54 studies across 21 countries and organizes the evidence around an explicit theory of change. We use design-appropriate tools to appraise each study and check a random sample of extracted data againts its original sources. Several limitations remain. Screening and quality appraisal were assisted by active learning, and extraction checks covered only a sample of the data. The resulting ratings represent a systematically applied assessment rather than a consensus judgment. Publication and language bias may also affect the evidence included.

5 Conclusion

This review examined whether horticulture improves food security and nutrition, the mechanisms through which these effects occur, and the intervention and policy features associated with better outcomes. We synthesised 54 empirical studies from 21 countries using a common extraction framework, appraised each study with a design-appropriate tool, and verified every extracted quantitative value against a located passage or table in the original source.

The answer is a qualified yes. Horticulture can improve food security and nutrition, but its effects are neither automatic nor uniform. Evidence is strongest for proximal outcomes—dietary diversity, fruit and vegetable consumption, and household food security—and weakest for nutritional status and anthropometric measures. This gradient may reflect measurement and timing rather than a limit to horticulture's nutritional potential. Diets can respond within a season, whereas growth and biomarker outcomes develop over longer periods and depend on health, sanitation, disease, and other non-food inputs. A literature dominated by short follow-up periods and farm-level outcomes is therefore more likely to detect dietary changes than anthropometric effects.

The evidence on mechanisms is highly concentrated. Production and food availability appear in 51 of the 54 studies, and income appears in 42. Agrobiodiversity, price transmission, and wider market spillovers receive little attention. This gap matters because these mechanisms operate beyond participating households and determine whether program-level benefits scale into broader market and system effects. The evidence is similarly uneven across the value chain: most studies focus on production, while post-harvest handling, storage, aggregation, and market-level interventions remain understudied despite substantial documented losses beyond the farm gate.

Program design also appears to matter. Effects tend to be larger when production support is combined with nutrition education, market access, or measures addressing women's control over resources and unpaid work. Increasing output alone does not determine whether households consume or sell the additional produce, whether it reaches markets before spoiling, or whether earnings are spent on nutritious food. Complementary program components can address these constraints. This conclusion remains tentative, however, because few studies include a production-only comparison group. The apparent advantage of bundled interventions therefore rests mainly on comparisons across studies rather than direct tests within the same evaluation.

These conclusions must be interpreted in light of the evidence's quality. Of the 54 studies, only 10 use randomized designs and 31 use designs capable of supporting causal claims; 9 have no comparison group, and 49 contain at least one documented problem in their reported statistics. **[TODO: quality-appraisal distribution pending — see the Methods note]** The direction of the evidence is therefore more credible than the precise magnitude of any individual estimate. Further progress requires not only more rigorous studies but also more consistent, transparent, and reproducible reporting.

Realising horticulture's potential for nutrition-sensitive development will require interventions tailored to their intended outcomes. Programs should combine production support with the complementary measures needed to translate greater output into better diets, protect household consumption and women's time as commercialization expands, and address constraints beyond the farm gate. Evaluations must also extend beyond proximal outcomes and short follow-up

periods, while reporting results clearly enough to permit independent verification and cumulative synthesis.

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Author contributions (CRediT). **[TODO: assign CRediT roles per author — [Author 3]: …; [Author 4]: …; [Author 5]: … (e.g. Conceptualization, Methodology, Formal analysis, Data curation, Writing — original draft, Writing — review & editing, Supervision)]**

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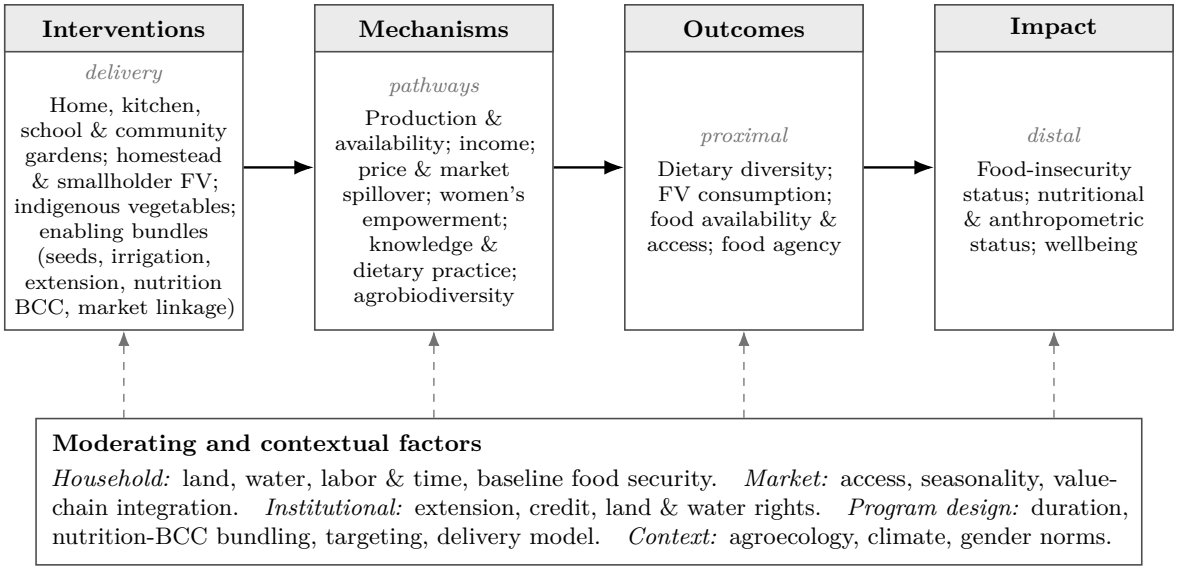
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A Appendix

A.1 Theory of change

Figure A1 presents the theory of change guiding the review, described in Section 2.1. Horticultural interventions act through a set of mechanisms on food-security and nutrition outcomes, sequenced from proximal to distal, and are moderated by household, market, institutional, program-design, and contextual factors that explain heterogeneity in effects.

Figure A1: Theory of change: horticultural interventions act through mechanisms on food-security and nutrition outcomes, moderated by household, market, institutional, program-design, and contextual factors. Adapted from the project framework.



A.2 Full search strategy

The same conceptual query was applied across Web of Science and Scopus, with syntax adapted to each database. Searches were restricted to English-language articles and review articles in the social-sciences domain, published 2005–2026. The Scopus `TITLE-ABS-KEY` query was:

```
(("horticulture" OR horticultural OR "horticultural crops" OR vegetable* OR fruit* OR
("home garden" OR "home gardens") OR ("kitchen garden" OR "kitchen gardens") OR ("homestead
garden" OR "homestead gardens") OR ("homestead production" OR "homestead food production")
OR ("school garden" OR "school gardens")) AND ("food security" OR "food insecurity" OR
nutrition OR "nutrition security" OR "dietary diversity" OR "food availability" OR "food
access" OR "food utilization" OR "food stability" OR "food sustainability" OR "food agency")
AND NOT ("ornamental horticulture" OR floriculture OR landscaping OR "cut flowers" OR
"food processing" OR "food industry"))
```

Applying the filters returned 7,759 records in Web of Science (from 31,287 unbounded) and 2,920 in Scopus (from 52,475 unbounded), or 10,679 in total; de-duplication in Rayyan removed 211 records, leaving 10,468 unique records.

A.3 Data extraction codebook

Each included study was charted against a standardized extraction template. Table A1 lists the charted fields and their coding instructions.

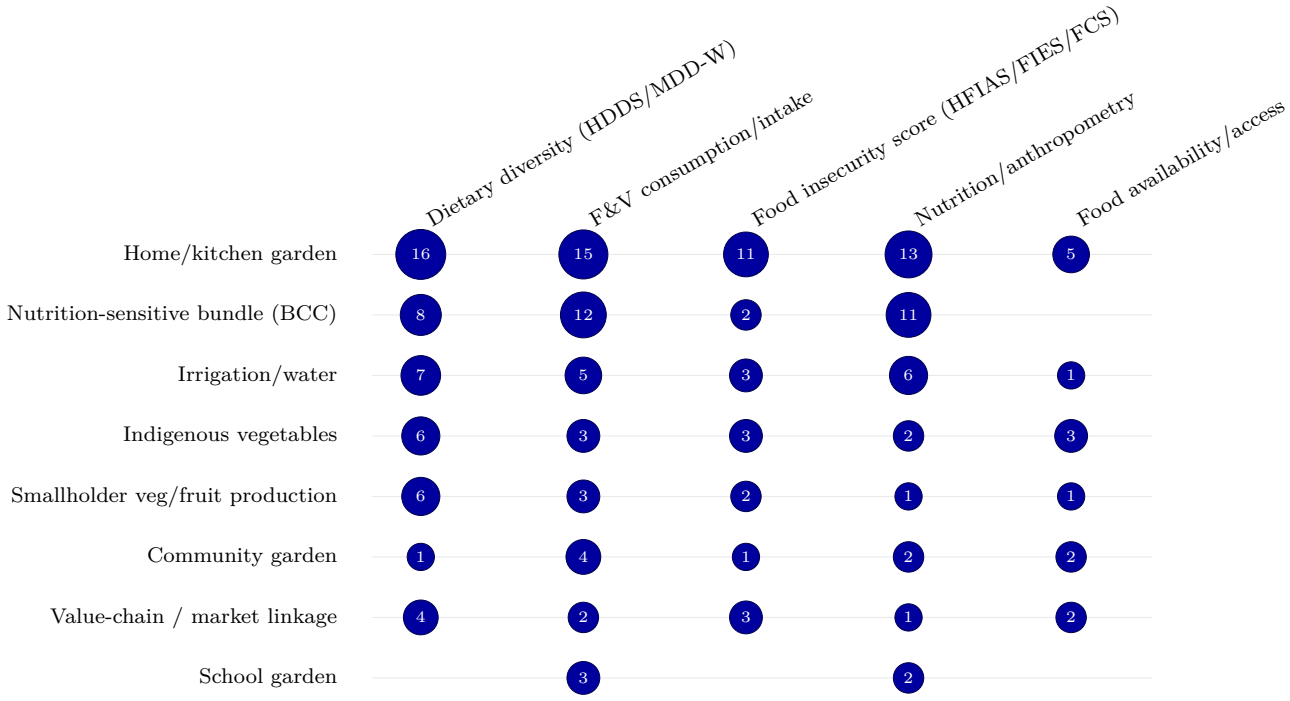
Table A1: Data-extraction codebook: charted fields and their coding instructions.

Field	**Definition / coding instructions**
Article ID	Sequential ID assigned by review team for tracking (A001, A002, ...).
Methodology Type	Quantitative / Mixed methods / Qualitative / Systematic review / RCT. Per team decision (09 Jun 2025 meeting), pure Qualitative-only primary studies are dropped unless they uniquely address an RQ with no quantitative alternative - record reasoning in Include/Drop Reason.
Include/Drop	Final screening decision at full-text/extraction stage.
Include/Drop Reason	Brief justification, esp. for borderline qualitative or mixed-methods calls.
Sample Size (n)	Total analytic sample size used for primary outcome(s).
Sample Size Breakdown	Sub-sample sizes by arm/subgroup/site where applicable.
Comparator Type	What the intervention/exposed group is compared against, if any.
Theoretical Framework	Named conceptual model used by the authors (e.g., Five A's, livelihoods/social capital, agrobiodiversity-functions, social-ecological model).
Food Security Framework/Instrument	Specific validated tool or approach used to measure food security/nutrition outcome (FCS, HFIAS, FIES, DDS/MDD-W, Five A's, anthropometric, self-report, etc.).
Outcome Direction by Sub-outcome	Record effect direction/significance separately for each outcome measured (a study can show effects on some sub-outcomes and not others - do not collapse to one overall verdict).
Heterogeneity/Subgroup Findings	Any reported variation in effect by wealth, gender, land ownership, geography, age, etc. Flag explicitly since several papers' main contribution is heterogeneity, not average effect.
Relevance to RQ1/RQ2/RQ3	Tag Yes/Limited/No plus 1-line justification for each of the team's 3 research questions to speed later synthesis.

A.4 Additional study-characteristics tables

This appendix reports the distribution of the 54 included studies by horticultural intervention type (Table A2), food-security and nutrition outcomes measured (Table A3), and impact pathways discussed (Table A4). Figure A2 cross-tabulates intervention type against outcome domain as an evidence-gap map.

Figure A2: Evidence-gap map of the included studies.



Note. Bubble area is proportional to the number of the 54 included studies measuring each food-security or nutrition outcome (columns) for each horticultural intervention type (rows); rows and columns are ordered by overall frequency. Tags are non-exclusive, so a study contributes to every applicable cell. Cell values are study counts.

Table A2: Horticultural intervention types (studies may be counted more than once).

Intervention type	**Studies (*n*)**	**%**
Home/kitchen garden	25	46%
Nutrition-sensitive bundle (BCC)	15	28%
Irrigation/water	8	15%
Indigenous vegetables	7	13%
Smallholder veg/fruit production	6	11%
Community garden	5	9%
Value-chain / market linkage	5	9%
School garden	3	6%

Note. Counts sum to more than 54 because many studies test bundled interventions.

Table A3: Food security and nutrition outcomes measured (multiple per study).

Outcome domain	**Studies (*n*)**	**%**
Dietary diversity (HDDS/MDD-W)	31	57%
F&V consumption/intake	31	57%
Food insecurity score (HFIAS/FIES/FCS)	21	39%
Nutrition/anthropometry	20	37%
Food availability/access	13	24%

Note. Non-exclusive; most studies report several outcomes.

Table A4: Impact pathways/mechanisms discussed (multiple per study).

Mechanism / pathway	**Studies (*n*)**	**%**
Production/availability	51	94%
Income	42	78%
Knowledge/dietary practice	29	54%
Women's empowerment	14	26%
Agrobiodiversity/diet diversity	2	4%
Price/market spillover	1	2%

Note. Coded from the mechanisms field; non-exclusive.

A.5 Evidence table

Table A5: Characteristics of the 54 included studies.

Study	**Country / Setting**	**Design**	**Outcomes**
Ayanwale et al. (2025)	Nigeria (LMIC)	Observational	HDDS
Baliki et al. (2024)	Syria (LMIC)	Quasi-experimental	FCS; F&V intake
Barnidge et al. (2014)	United States (HIC)	Quasi-experimental	F&V intake
Blakstad et al. (2022)	Tanzania (LMIC)	RCT	HFIAS; MDD-W; F&V intake; Anthropometry; ...
Bliznashka et al. (2022)	Burkina Faso (LMIC)	RCT	Anthropometry
Burt et al. (2021)	Multi-country: primarily... (Mixed)	Systematic review	F&V intake; Food availability/access
Bushamuka et al. (2005)	Bangladesh (LMIC)	Observational	HFIAS; FCS; DDS; F&V intake; ...
Castañeda-Navarrete (2021)	Mexico (LMIC)	Observational	FCS; F&V intake

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Table A5 continued

Study	Country / Setting	Design	Outcomes
Cheng et al. (2021)	Marshall Islands (LMIC)	Mixed methods	DDS; F&V intake; Anthropometry; Food availability/access
Danso-Abbeam et al. (2021)	Rwanda (LMIC)	Quasi-experimental	FIES
Davis et al. (2021)	United States (HIC)	RCT	F&V intake; Anthropometry
Depenbusch et al. (2021)	Kenya; Tanzania; Uganda (LMIC)	RCT	HDDS; Food availability/access
Depenbusch et al. (2022)	Cambodia (LMIC)	RCT	HDDS; Micronutrient status
D'Haese et al. (2013)	South Africa (LMIC)	Observational	HFIAS; HDDS; Food availability/access
Dragojlovic et al. (2020)	Cambodia (LMIC)	Quantitative (other)	F&V intake; Micronutrient status
Fehr and Moseley (2019)	Botswana (LMIC)	Mixed methods	HFIAS; DDS; Food availability/access
Gallaher et al. (2013)	Kenya (LMIC)	Mixed methods	HDDS; F&V intake
Garcia et al. (2018)	Multiple countries:... (Mixed)	Systematic review	F&V intake; Anthropometry; Food availability/access
Gebru et al. (2019)	Ethiopia (LMIC)	Mixed methods	HFIAS; MAHFP; F&V intake; Anthropometry; ...
Gwacela et al. (2024)	South Africa (LMIC)	Quasi-experimental	HFIAS; Food availability/access
Issahaku et al. (2023)	Rwanda (LMIC)	Quasi-experimental	FCS; HDDS; F&V intake; Anthropometry; ...
Jones et al. (2005)	Nepal (LMIC)	Observational	F&V intake; Micronutrient status
Kabunga et al. (2014)	Kenya (LMIC)	Quasi-experimental	HFIAS; F&V intake; Food availability/access
Kahsay et al. (2025)	Ethiopia (LMIC)	Quasi-experimental	HDDS; F&V intake
Kalavathi et al. (2010)	India (LMIC)	Quasi-experimental	F&V intake
Keding et al. (2012)	Tanzania (LMIC)	Observational	DDS
Kone et al. (2025)	Côte d'Ivoire (LMIC)	Observational	FCS; DDS; F&V intake
Krause et al. (2019)	Kenya (LMIC)	Quasi-experimental	FCS; MAHFP; CSI; HDDS; ...
Kubitza et al. (2024)	Kenya (LMIC)	Mixed methods	HDDS; F&V intake
Kubuga (2025)	Ghana (LMIC)	Quasi-experimental	MDD-W

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Table A5 continued

Study	Country / Setting	Design	Outcomes
Lin et al. (2024)	United States (HIC)	Observational	Food availability/access
Malatsi et al. (2023)	South Africa (LMIC)	Observational	Food availability/access
Mossie and Elias (2025)	Ethiopia (LMIC)	Mixed methods	HDDS
Ndwandwe et al. (2026b)	Sub-Saharan Africa (Kenya 9… (LMIC)	Systematic review	HDDS; Micronutrient status; Food availability/access
Ndwandwe et al. (2026a)	Sub-Saharan Africa… (LMIC)	Systematic review	DDS; Micronutrient status; Food availability/access
Nkomoki et al. (2025)	Ghana (LMIC)	Quasi-experimental	HFIAS; Food availability/access
Nyang'au et al. (2020)	Kenya (LMIC)	Quasi-experimental	HDDS; F&V intake; Food availability/access
Ogunleye et al. (2025)	Lesotho (LMIC)	Quasi-experimental	FIES
Ogutu et al. (2023)	India (LMIC)	Quasi-experimental	HDDS; MDD-W; Anthropometry
Olney et al. (2009)	Cambodia (LMIC)	Quasi-experimental	HDDS; F&V intake; Anthropometry; Micronutrient status
Osei et al. (2017)	Nepal (LMIC)	RCT	MDD-W; Anthropometry; Micronutrient status; Food availability/access
Prah et al. (2026)	Ghana (LMIC)	Quasi-experimental	HFIAS; HDDS; Food availability/access
Rahman et al. (2024)	Bangladesh (LMIC)	Quasi-experimental	HDDS; F&V intake
Rammohan et al. (2019)	Myanmar (LMIC)	Observational	HFIAS; HDDS; MDD-W; Food availability/access
Sanchez et al. (2025)	United States (HIC)	RCT	F&V intake
Schreinemachers et al. (2015)	Bangladesh (LMIC)	Quasi-experimental	Diversity of vegetables consumed (count of different…
Schreinemachers et al. (2016)	Bangladesh (LMIC)	Quasi-experimental	DDS; F&V intake
Schreinemachers et al. (2017)	Nepal (LMIC)	RCT	F&V intake; Anthropometry
Schreinemachers et al. (2019)	Burkina Faso (LMIC)	RCT	F&V intake; Anthropometry
Schreinemachers et al. (2025)	Bangladesh (LMIC)	Quasi-experimental	FIES; MDD-W; F&V intake
Tafesse et al. (2020)	Ethiopia (LMIC)	Quasi-experimental	FCS; DDS; F&V intake; Anthropometry; …

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Table A5 continued

Study	Country / Setting	Design	Outcomes
Taylor et al. (2018)	United States (HIC)	RCT	F&V intake; Anthropometry; Food availability/access
Whitney et al. (2018)	Uganda (LMIC)	Observational	HFIAS; HDDS; MDD-W; Anthropometry; ...
Zondi et al. (2022)	South Africa (LMIC)	Observational	HFIAS; Food availability/access